An atomic-boundary minimizer with no closed representative

A counterexample to Question 5.2 of arXiv:2305.02891

Autonomous proof attempt, lane 6

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Status: candidate full counterexample, likely valid. Question 5.2 has a negative answer for both functionals in the general class of metric measure spaces stated by the source.

1 The source question

For a bounded domain Ω in a metric measure space $(X, d, \mathfrak{m})$, Koivu, Lučić, and Rajala consider

$$M_\lambda(A) = \mathrm{Per}(A) + \lambda \mathfrak{m}(\Omega \setminus A) \quad (A \subset \Omega)$$

and

$$\widetilde{M}_\lambda(A) = \mathrm{Per}(A) + \lambda \mathfrak{m}(\Omega \triangle A) \quad (A \subset X).$$

Their Question 5.2 asks whether a Borel minimizer of either functional must have a closed representative [1, Question 5.2]. Here *representative* means equality up to an $\mathfrak{m}$-null set. The exact question appears on PDF page 17 of the source and is reproduced in Figure 1.

2 Proof intuition

Put positive atoms exactly at the two boundary points of an interval. Because convergence in $L^1(\mathfrak{m})$ detects the atoms, every Lipschitz approximation to the indicator of the open interval must be near zero at both endpoints and near one somewhere inside. It therefore pays at least one unit of variation going up and one going down. Thus the open interval has perimeter two and no non-null competitor can have smaller perimeter.

For a fidelity weight $\lambda \geq 2$, deleting the interval costs at least λ, while including either endpoint in the symmetric-difference problem also costs at least λ. The open interval is consequently a minimizer of both functionals. But a closed set equal to the interval almost everywhere would have to exclude the two positive-mass endpoints; closedness then forces it to omit an entire one-sided neighborhood, of positive Lebesgue measure. Hence no closed representative exists.

3 The counterexample

Let

$$X = \mathbb{R}, \qquad d(s,t) = |s - t|, \qquad \mathfrak{m} = \mathcal{L}^1 + \delta_0 + \delta_1, \qquad \Omega = (0,1).$$

This is a complete separable metric measure space, its Borel measure is finite on bounded sets and has full support, and Ω is a bounded domain. We write $\mathrm{Per}(A) = |D\chi_A|(X)$ for the relaxed metric perimeter used in the source.

Lemma 1. *1.* $\mathrm{Per}(\Omega) = 2$.

2. If $A \subset X$ is Borel, $0, 1 \notin A$, and $\mathcal{L}^1(A \cap (0,1)) > 0$, then $\mathrm{Per}(A) \geq 2$.

Proof. For the upper bound in (1), let $0 < \varepsilon < 1/4$ and choose the piecewise linear function u_ε which is zero outside $(\varepsilon, 1-\varepsilon)$, rises linearly from zero to one on $[\varepsilon, 2\varepsilon]$, equals one on $[2\varepsilon, 1-2\varepsilon]$, and falls linearly to zero on $[1-2\varepsilon, 1-\varepsilon]$. Then $u_\varepsilon \to \chi_\Omega$ in $L^1(\mathfrak{m})$. It is constant near both atoms, so their contributions to $\int_X \mathrm{lip}_a(u_\varepsilon)\, d\mathfrak{m}$ vanish, while the Lebesgue contribution is exactly two. Hence $\mathrm{Per}(\Omega) \leq 2$.

For the lower bound, consider any Lipschitz relaxation sequence u_j for χ_A. Composing with the truncation onto $[0,1]$ does not increase the asymptotic slope, so assume $0 \leq u_j \leq 1$. Since the measure has atoms at 0 and 1 and these points are not in A,

$$u_j(0) \longrightarrow 0, \qquad u_j(1) \longrightarrow 0. \tag{1}$$

If $\mathcal{L}^1(A \cap (0,1)) > 0$, convergence in the Lebesgue part of $L^1(\mathfrak{m})$ gives points $t_j \in A \cap (0,1)$ for which $u_j(t_j) \to 1$. The fundamental theorem for Lipschitz functions and the inequality $|u_j'| \leq \mathrm{lip}_a(u_j)$ almost everywhere give

$$\begin{aligned}\int_X \mathrm{lip}_a(u_j)\, d\mathfrak{m} &\geq \int_0^1 \mathrm{lip}_a(u_j)\, dx \\ &\geq |u_j(t_j) - u_j(0)| + |u_j(1) - u_j(t_j)|.\end{aligned}$$

Taking the lower limit and using (1) proves $\mathrm{Per}(A) \geq 2$. Applying this to $A = \Omega$ gives the reverse inequality in (1). $\square$

Theorem 2. *For every $\lambda \geq 2$, the Borel set $E = \Omega$ is a minimizer of both M_λ among Borel subsets of Ω and $\widetilde{M}_\lambda$ among Borel subsets of X. Nevertheless, E has no closed representative. Therefore Question 5.2 of [1] has a negative answer for both functionals.*

Proof. First consider M_λ. If $A \subset \Omega$ and $\mathcal{L}^1(A) > 0$, then Lemma 1 gives

$$M_\lambda(A) \geq \mathrm{Per}(A) \geq 2.$$

If $\mathcal{L}^1(A) = 0$, then A is $\mathfrak{m}$-null, because neither endpoint belongs to Ω. Consequently

$$M_\lambda(A) = \lambda \mathfrak{m}(\Omega) = \lambda \geq 2.$$

On the other hand, $M_\lambda(\Omega) = \mathrm{Per}(\Omega) = 2$, so Ω is a minimizer.

Now consider an arbitrary Borel $A \subset X$ for $\widetilde{M}_\lambda$. If $0 \in A$ or $1 \in A$, then the corresponding unit atom lies in $\Omega \triangle A$, and hence

$$\widetilde{M}_\lambda(A) \geq \lambda \geq 2.$$

If neither endpoint lies in A and $\mathcal{L}^1(A \cap \Omega) > 0$, then Lemma 1 again yields $\widetilde{M}_\lambda(A) \geq 2$. In the remaining case $0, 1 \notin A$ and $\mathcal{L}^1(A \cap \Omega) = 0$, the whole Lebesgue mass of Ω is missing, so

$$\widetilde{M}_\lambda(A) \geq \lambda \mathfrak{m}(\Omega \setminus A) = \lambda \geq 2.$$

Since $\widetilde{M}_\lambda(\Omega) = 2$, the same $E = \Omega$ minimizes the second functional.

Suppose finally that a closed $F \subset X$ were a representative of Ω. Equality modulo $\mathfrak{m}$ forces $0, 1 \notin F$, since both points have positive atomic mass and lie outside Ω. Since F is closed and $0 \notin F$, some interval $(-r, r)$ is disjoint from F. Then $(0, r) \subset \Omega \setminus F$ has positive Lebesgue measure, contradicting $\mathfrak{m}(F \triangle \Omega) = 0$. Thus no closed representative exists. $\square$

4 Verification and scope

The proof uses only the source definitions. The lower perimeter bound is insensitive to behavior outside $[0, 1]$: endpoint atoms force the two endpoint values in every relaxation sequence, and one interior point with value approaching one forces total variation at least two. The explicit tent sequence proves sharpness and has zero asymptotic slope at the atoms.

The example is deliberately outside the PI setting. This is necessary for the $\widetilde{M}_\lambda$ part, because the source already proves that $\widetilde{M}_\lambda$ minimizers have closed representatives in PI-spaces. Question 5.2, however, is stated for arbitrary metric measure spaces, exactly the class containing the example.

A bounded novelty search covered the run indexes and the exact wording of Question 5.2. It also covered arXiv:2305.02891, its published IMRN version, and the later arXiv:2503.15716 paper by overlapping authors. No source resolving Question 5.2 was found. The IMRN version still labels it open. This is a bounded search, not a definitive novelty certification.

Human-review recommendation. Review as a likely valid full counterexample. The central check is the relaxed-perimeter lower bound in Lemma 1; all minimization and representative claims then follow by a three-case comparison.

Question 5.2. Let $(\mathrm{X}, \mathsf{d}, \mathfrak{m})$ be a metirc measure space and $\Omega \subset \mathrm{X}$ a bounded domain. Let E be a minimizer of M_λ (or $\widetilde{M}_\lambda$) among Borel subsets of Ω (or X respectively). Does E have a closed representative?

For PI-spaces the answer to Question 5.2 is positive for $\widetilde{M}_\lambda$, see again Section 4.

Independent of the minimization approach, the obvious question still remaining is:

Question 5.3. Let $(\mathrm{X}, \mathsf{d}, \mathfrak{m})$ be a metric measure space, $\Omega \subset \mathrm{X}$ a bounded domain and $\varepsilon > 0$. Does there exist a BV-extension domain $A \subset \Omega$ such that $\mathfrak{m}(\Omega \setminus A) < \varepsilon$?

None of our approximations is from outside because we argue that the minimizer is an extension set by comparing the value of the functional to value at a modification of the minimizer where we take away an open subset.

Question 5.4. Let $(\mathrm{X}, \mathsf{d}, \mathfrak{m})$ be a metric measure space, $\Omega \subset \mathrm{X}$ a bounded domain and $\varepsilon > 0$. Does there exist a BV-extension domain (or just a BV-extension set) $A \supset \Omega$ such that $\mathfrak{m}(A \setminus \Omega) < \varepsilon$?

In addition to knowing the answer to the above questions, it would be interesting to see if we can also approximate domains by Sobolev $W^{1,p}$-extension domains in the absence of the local Poincaré inequality. In particular, the case $p = 1$ is intimately connected to the BV and perimeter extensions even in general metric measure spaces [8].

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Figure 1: Question 5.2 on PDF page 17 of arXiv:2305.02891.

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